

A rain-on-grid flood model of the Al-Masar Al-Akhdar corridor

What the simulator is, what it does, and how it performs

This note describes a two-dimensional hydraulic model of pluvial (rainfall) flooding built for the Fouad Boutros right-of-way in Beirut, and the results obtained from it. It covers the governing equations and how they are solved, how the terrain and the proposed green corridor are represented, the verification and validation evidence, and the findings for the corridor itself. Numbers quoted throughout are read directly from the model outputs.

1. What the model solves

The model integrates the two-dimensional shallow-water equations — depth-averaged conservation of mass and momentum — over a regular grid. In conservative form,

$$\frac{\partial}{\partial t} \begin{pmatrix} h \\ hu \\ hv \end{pmatrix} + \frac{\partial}{\partial x} \begin{pmatrix} hu \\ hu^2 + \frac{1}{2}gh^2 \\ huv \end{pmatrix} + \frac{\partial}{\partial y} \begin{pmatrix} hv \\ huv \\ hv^2 + \frac{1}{2}gh^2 \end{pmatrix} = \begin{pmatrix} R - I - D \\ -gh \partial_x z_b - \tau_x \\ -gh \partial_y z_b - \tau_y \end{pmatrix}$$

where h is water depth, (u, v) the depth-averaged velocity, z_b the bed elevation, and τ the bed friction. The three source terms on the mass equation are what make this a *pluvial* model rather than a river model: R is rainfall applied directly to every cell, I is infiltration into the ground, and D is capture by storm-drain inlets.

Rain-on-grid, and why it matters here

Conventional flood studies delineate sub-catchments, estimate a runoff coefficient for each, and route the resulting hydrographs through a drainage network. That approach requires the modeller to decide in advance where the water goes. Here, rainfall is instead applied to every cell of the grid and the water finds its own path downhill under the momentum equations. The flood pattern is therefore an *output* of the model rather than an input to it, which is the property that lets it evaluate an intervention — such as a green corridor — whose whole purpose is to change where the water goes.

Two numerical schemes

The equations are solved on a Cartesian grid in `torch`, on GPU, with two selectable flux schemes:

- **Inertial (three-term).** The formulation of Bates, Horritt & Fewtrell (2010), which neglects the advective acceleration term and updates face-centred discharge semi-implicitly against Manning friction. Fast and widely used — it is the basis of LISFLOOD-FP — but it smears sharp flood fronts.
- **Shock-capturing (HLLC).** A Godunov finite-volume scheme with an HLLC approximate Riemann solver, Audusse hydrostatic reconstruction for well-balancedness over an irregular bed, and Kurganov–Petrova velocity desingularisation at wet/dry fronts. Roughly $3.5\times$ the cost of the inertial scheme, and considerably more accurate here.

Both schemes share the same terrain, source terms and mass accounting, so the difference between them isolates the effect of the numerics — a comparison exploited in Section 6.

Surface and sub-surface processes

Beyond the hydraulics, each cell carries its own surface properties, derived from a nine-class land-cover classification of the LiDAR returns and aerial imagery:

- **Infiltration**, either at a fixed rate or — the preferred option — by the Green–Ampt wetting-front model, $f = K(1 + \psi\Delta\theta/F)$, where K is saturated conductivity, ψ the wetting-front suction head, $\Delta\theta$ the initial moisture deficit and F the cumulative infiltration. This is solved implicitly over each timestep from the cumulative form $F - F_n - \psi\Delta\theta \ln[(F + \psi\Delta\theta)/(F_n + \psi\Delta\theta)] = K\Delta t$, because the explicit rate is unbounded on dry ground at the start of a storm. Infiltration capacity therefore *declines* as the ground wets, which a fixed rate cannot express.
- **Surface detention**, a per-cell depression depth, so that engineered features such as bioswales and bioretention ponds hold water rather than shedding it.
- **Storm-drain inlets**, either as fixed-capacity sinks or with a head–discharge relation — weir control while the grate is free-flowing, $Q = C_w Ph^{3/2}$, and orifice control once drowned, $Q = C_o A \sqrt{2gh}$, taking whichever is smaller. Capture then depends on the ponding the design actually produces.
- **Roof and courtyard routing**. Buildings stand as obstacles at roof height; rainfall landing on them, and in enclosed courtyards, is transferred to the nearest street cell, as a downspout would.
- **An opt-in erosion sub-model**, in which erodible soft surfaces degrade under sustained flow, lowering their live infiltration and roughness.

Mass conservation is verified on every run. The model reports the full balance — rainfall in, versus infiltration plus drainage plus outflow plus stored volume — and every result in this note closes it to better than 3×10^{-5} relative error. This matters because a model that loses or creates water silently can produce almost any answer; one of the three comparison engines below fails exactly this test.

2. How the terrain is built

The model domain is a $571 \text{ m} \times 1000 \text{ m}$ corridor strip together with its upslope catchment, gridded at 0.5 m into 2.3 million cells, from a 39 GB airborne LiDAR survey of Beirut.

The processing chain streams the point cloud once to grid lowest and highest returns per cell, then: fills holes and despeckles the ground surface; classifies land cover using excess-green, surface relief, slope and local roughness; rasterises buildings and water from OpenStreetMap; removes tree canopy and re-interpolates the ground beneath it; detects enclosed courtyards; builds the rain-routing raster; and derives the Manning, infiltration and soil-parameter grids.

Two terrain corrections made during this work are worth recording, because both changed results materially:

- **A non-physical excavation.** A $1,988 \text{ m}^2$ patch read -0.8 m — some 27 m below sea level — while the ground enclosing it sat at 37 m. Photogrammetry has no reliable floor for a dark, enclosed excavation. It had been masked as outflow water, which silently deleted every drop of runoff reaching it. It is now identified and modelled as a bounded basin.
- **Building demolition.** The corridor clears buildings in its path. The fill used to replace them drew elevations from the nearest “open ground”, but the OpenStreetMap building extract is

incomplete over this domain, so unmapped rooftops were being treated as ground: 14 of 35 cleared footprints ended up still standing more than 3 m above their surroundings, the worst by 23 m. Fill sources are now screened by a morphological top-hat filter that rejects anything standing like a roof, whether or not OSM knows about it.

3. How the green corridor is represented

The corridor is not modelled as a polygon with an average infiltration rate. Its geometry comes from the AUB Beirut Urban Lab dataset — the 19 ha Green Path zone polygon and the Fouad Boutros right-of-way alignment — and the designed cross-section is written into the terrain cell by cell, each of seven surface types carrying its own infiltration rate, Manning roughness and detention depth.

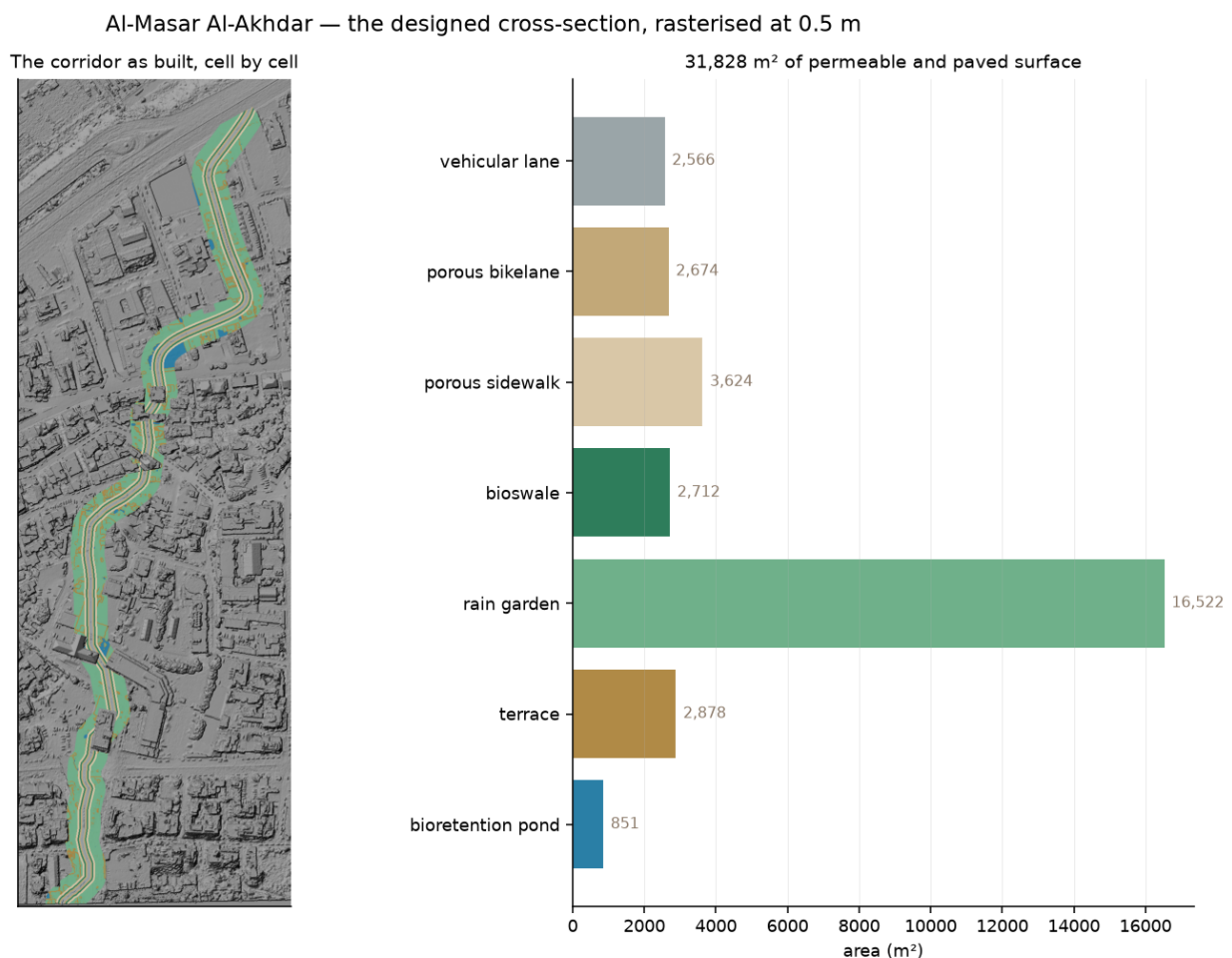


Figure 1. The corridor as modelled, and its composition by area. Rain garden accounts for just over half of the 31,828 m²; the engineered bioswales and bioretention ponds together come to under 4,000 m².

The internal layout is an interpretation. The zone polygon and the right-of-way alignment are data. The cross-section — band widths, where the bioswale sits relative to the sidewalk, where ponds and terraces are placed — was read from the published design drawings and completed with what appeared hydraulically sensible. Ponds are sited at terrain low points and terraces on the steep segments, which is standard practice and also where the model independently shows water collecting. This is faithful in spirit but is not a construction specification; supplied dimensions could be substituted directly, as the pipeline consumes a material raster.

Because the corridor also demolishes buildings, the clearing is applied as a separate terrain layer that the corridor design switches on. The “before” case therefore retains every building at full height, and building the corridor is what removes those in its path — so the before/after comparison is a genuine one rather than two views of an already-cleared site.

4. Verification: does it solve the equations correctly?

Five analytic tests, where the correct answer is known exactly, are run after any change to the solver. All five pass on both schemes.

Test	What it checks	Result
Closed box	Volume conservation, no leakage over walls	exact
Planar runoff	Steady depth against Manning normal depth	14.1 vs 14.7 mm
Lake at rest	No spurious currents over an uneven bed	6×10^{-5} m/s
Port equivalence	Agreement with an independent implementation	0.027 mm
Erosion sanity	Inert when disabled; mass still closes when on	bit-identical

The lake-at-rest test deserves comment: it is the check that a scheme is *well-balanced*, meaning it can represent still water over an irregular bed without generating artificial flow. Many schemes fail it, and a failure is serious in a domain like this one, which is terraced with retaining walls.

5. Validation: does it agree with reality, and with others?

Independent solver

The strongest available evidence is a comparison against SynxFlow — a separately written full shallow-water Godunov code — on the identical grid and storm.

Our scheme	Extent IoU	Depth correlation	RMSE	Wet-area bias
Inertial (three-term)	0.690	0.938	7.8 cm	+27%
Shock-capturing (HLLC)	0.913	0.985	3.7 cm	+1%

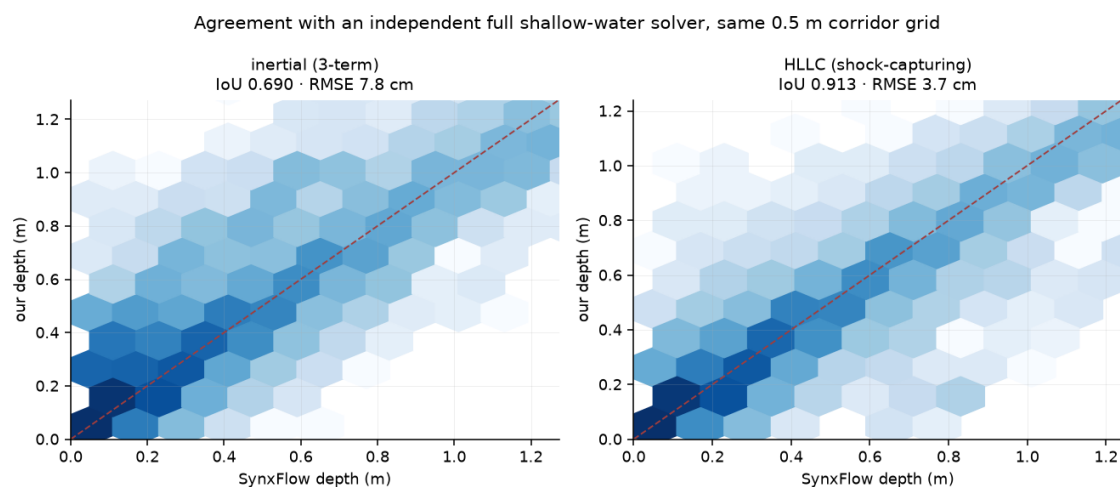


Figure 2. Cell-by-cell depth agreement with SynxFlow. Two independently written Godunov schemes agree to 3.7 cm across a full urban catchment.

Two conclusions follow. First, the shock-capturing scheme is corroborated about as far as model-to-model evidence permits. Second, the simplified inertial scheme carries a systematic bias here: it spreads water over 27% more area than the full solver, which is the expected consequence of neglecting advective momentum on streets that run supercritical at 6–8 m/s on a 6% grade.

A third engine, LISFLOOD-FP, could not be used at all. On this stepped terrain it generated roughly six times the input water volume and failed its own mass-balance check, so its output is void rather than merely different. This is a documented limitation of its scheme on terrain with retaining walls.

International benchmarks

Two cases from the UK Environment Agency benchmark suite (Néelz & Pender, report SC120002), which publishes results from 19 commercial and research packages.

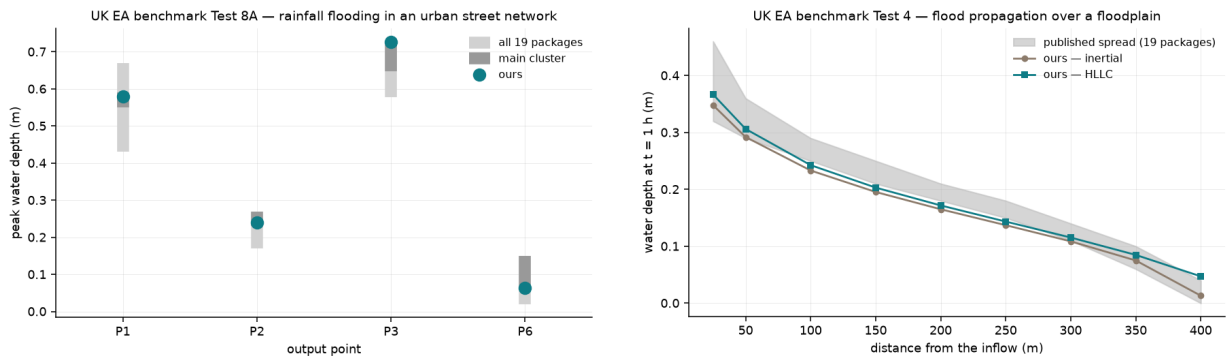


Figure 3. Left: Test 8A, rainfall flooding in an urban street network — inside the main published cluster at all four gauges with published curves. Right: Test 4, flood propagation over a floodplain — inside the published spread at all nine cross-section points, on both schemes.

Test 8A is the closest published analogue to this application: direct rainfall on a real, dense street network. Test 4 tests the celerity of an advancing front. Of the remaining six benchmark cases, four require terrain and boundary data not available to us, and two require coupled one-dimensional sewer or river models, which this tool does not implement.

What is *not* validated. Every comparison above is against another model or a benchmark. There is no observed flood data for Beirut — rainfall for the 25 November 2025 event is recorded, but no measured depths, extents or timings anywhere in the study area. The model is therefore *verified* but not *validated* in the strict sense. Closing this requires field measurement: levelled water marks at a few dozen street locations after a significant storm, and ideally a small number of depth loggers at known ponding points.

6. Grid resolution: does half a metre earn its cost?

The EA benchmark report states that “it is not currently clear that grid resolutions finer than 2 m will improve the quality of velocity predictions”, because terrain error and boundary conditions act at the same order as the grid, and recommends the question be tested at 0.5 m. We are unusually well placed to answer it. The same terrain was coarsened by block aggregation, so that the computational grid is the only quantity varying.

Resolution (m)	Cells	Corridor benefit	p99 street velocity
0.5	2 285 142	9.4	4.3
1.0	571 000	2.4	8.1
2.0	142 500	0.6	11.9
4.0	35 500	2.4	14.8

At 2 m — the resolution the benchmark itself uses — the measured benefit of the corridor falls to 0.6%, from 9.4% at 0.5 m. A study conducted at that resolution would conclude the intervention does essentially nothing. Peak street velocity meanwhile rises as the grid coarsens, which is both the wrong direction and physically implausible: block-averaging manufactures artificial steep gradients between cells while erasing the kerbs and narrow streets that actually convey the water.

For appraising interventions of this kind, sub-metre resolution is therefore not a refinement but a precondition. The caveat is that coarsening an existing terrain is not identical to building a model natively at 2 m, where roughness would normally be recalibrated; the direction and magnitude of the effect are nonetheless unambiguous.

7. Results for the corridor

Three design storms are used: a frequent 2-year event, the observed 25 November 2025 storm (25.4 mm in 30 minutes — by the Lebanese stormwater code’s own intensity table only a 2–5 year burst, yet it flooded Sassine Square and the Ring), and a severe 50-year event.

Al-Masar green corridor — 25 Nov 2025 observed storm, 0.5 m grid

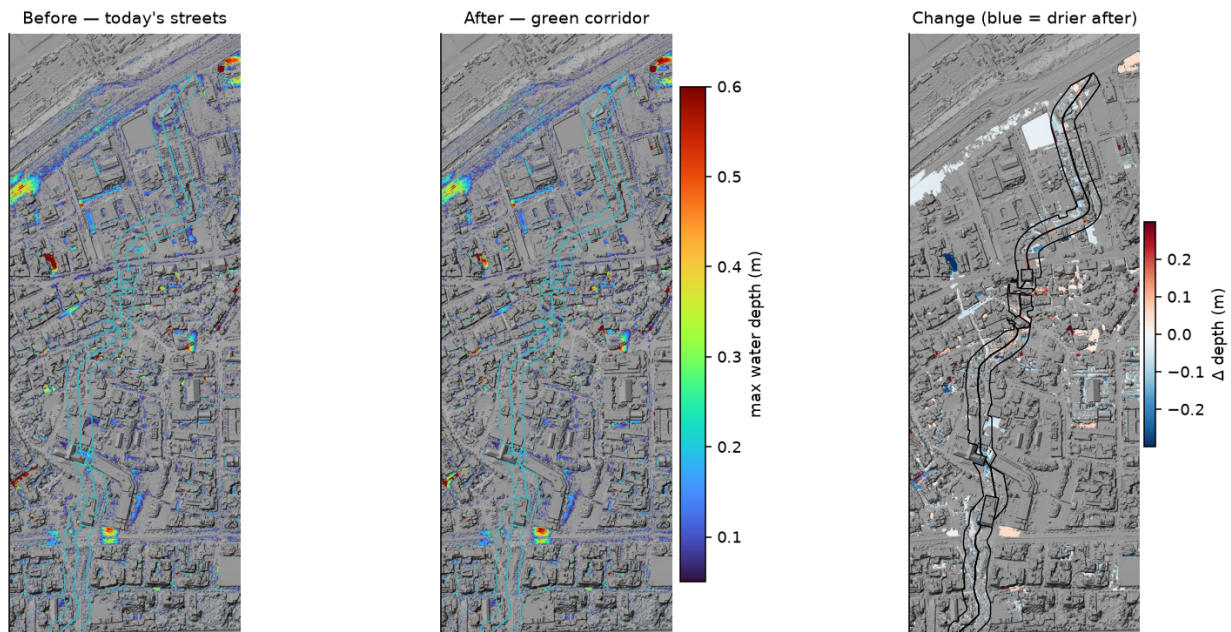


Figure 4. Peak water depth during the observed storm, before and after the corridor, with the difference at right. Blue indicates a drier outcome. The effect extends beyond the right-of-way into the surrounding street network, because the corridor intercepts runoff descending from Achrafieh.

Flooded-area reduction

Reductions are quoted on the corridor’s *walkable* surface. Approximately a fifth of the corridor is deliberately excavated as bioswales, rain gardens and terraces, which hold water by design; counting a full bioretention pond as “flooding” would penalise the design for functioning. Those

cells are 20% of the corridor by area but 34% of its wet area, so the distinction is material.

Storm	Baseline	+ shock capturing	+ Green–Ampt	Net change
T2 frequent	55.6	52.9	58.1	2.5
25 Nov 2025 observed	44.3	42.8	52.0	7.6
T50 severe	26.4	-9.0	-0.7	-27.1

Reduction in flooded walkable area (%), by model configuration. The final column is the difference between the best-validated configuration and the baseline.

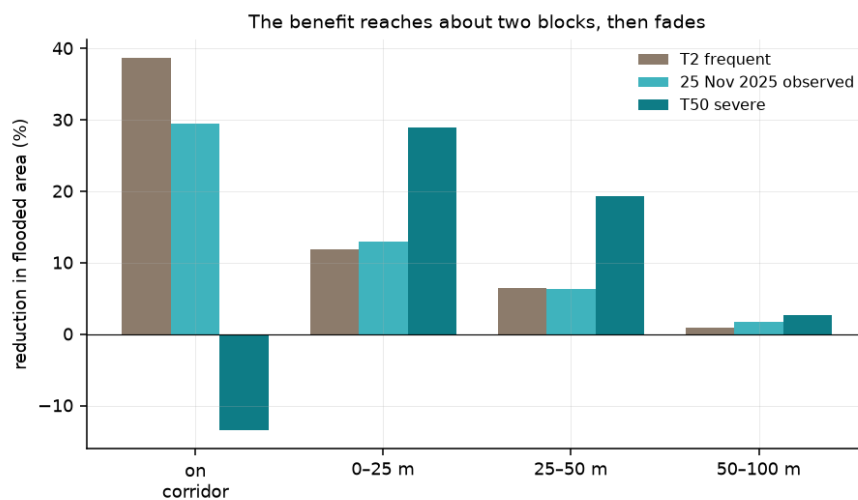


Figure 5. Reduction by distance from the corridor. The benefit extends roughly two blocks before fading — a useful planning figure, and a reminder that this is a linear intervention with a local catchment rather than a city-wide remedy.

Water balance

Storm	Infil. before (m ³)	after (m ³)	Gain (m ³)	Multiple	Outflow cut (%)
T2 frequent	3733	4644	+911	1.24	18
25 Nov 2025 observed	2898	3817	+919	1.32	17
T50 severe	4640	6765	+2124	1.46	11

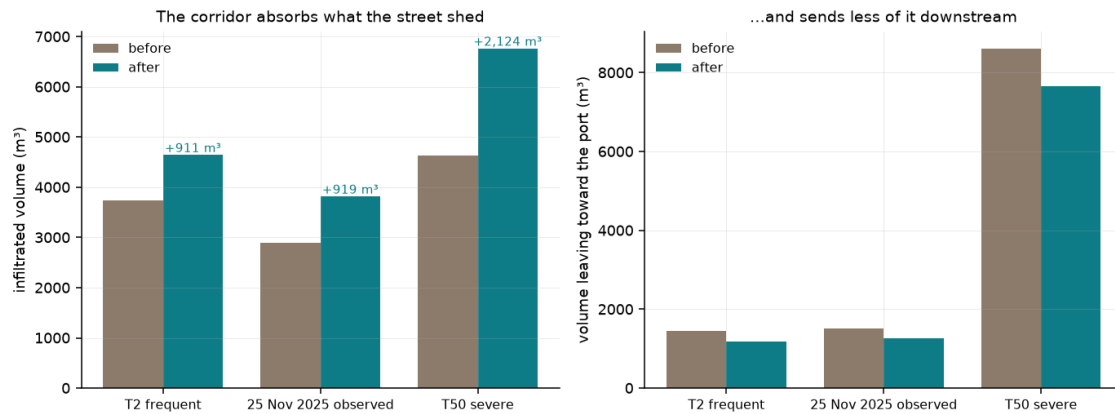


Figure 6. Infiltrated volume and volume discharged toward the port, before and after. The corridor absorbs more than the rain falling on it, because it also intercepts runoff from upslope.

Two findings that revise the earlier study

The severe storm. In a 50-year event the corridor no longer reduces the flooded area on its own footprint. It continues to perform hydrological work — absorbing over 2,000 m³ more water and cutting the volume discharged toward the port by around a tenth — but its swales and ponds fill and hold, so the water stands on the corridor by design instead of running down the surrounding streets. The corridor therefore has two distinct functions, keeping its own surface usable and protecting what lies downstream, and only the second survives an extreme event.

The attribution table above shows this is a *numerical* result rather than a soil-physics one: shock capturing alone moves the severe-storm figure by –35 percentage points, against 1–3 points for the two lighter storms. The earlier scheme was over-spreading water in the before-case, where flow is fastest and most supercritical, and thereby inflating the apparent benefit.

Infiltration. An earlier version of this study credited the corridor with roughly tripling infiltration. Under the Green–Ampt model the multiple is about 1.3. The absolute gain holds — some 900 m³ in the observed storm, within the previously published range — but ordinary Beirut ground absorbs considerably more than a fixed-rate model implied, so the corridor’s *relative* contribution is smaller. Better physics, smaller claim.

8. Limitations

- **No observed flood data.** The single largest gap; see Section 5.
- **Material properties are literature values** — bioretention and permeable-pavement infiltration rates from published guidance, taken conservatively — not site measurements. Nothing is calibrated to Beirut soils. The Green–Ampt soil parameters add four uncalibrated values per cover class.
- **Drain inlet capacity is assumed;** no per-inlet data exists. Results are reported as differences, which are robust to a common bias.
- **The corridor cross-section is interpreted** from design drawings, not specified.
- **The erosion sub-model is illustrative,** not calibrated, and is disabled in all results quoted here.
- **No coupled one-dimensional sewer or river model,** which places two EA benchmark cases out of scope.

9. What is distinctive about this model

Set against the 19 packages in the EA benchmark comparison, two differences are worth noting for any future publication.

None of those packages models infiltration or green infrastructure at all. They are hydraulics-only: shallow-water equations plus Manning roughness. The per-cell infiltration, detention and head-dependent inlet capture here make this a tool for appraising nature-based interventions rather than only for mapping floods.

None of them is differentiable. Because the solver is written in `torch`, the gradient of a flood objective with respect to every cell of a design can be obtained by automatic differentiation — verified against central finite differences to 2×10^{-8} , with gradient checkpointing making full-length storms tractable. This permits gradient-based design optimisation rather than trial of a handful of candidate layouts, and is the most promising direction for publication. Initial experiments are encouraging but not yet conclusive: the continuous optimum outperforms both a uniform and a greedy-heuristic design at equal area, but does not survive projection onto a buildable binary layout, which indicates the relaxation needs a projection scheme before the result can be claimed.

Model: 2.3 M cells at 0.5 m, GPU shallow-water, mass balance closing to $\sim 10^{-5}$. Terrain from a 39 GB airborne LiDAR survey; corridor geometry from the AUB Beirut Urban Lab Al-Masar Al-Akhdar dataset. All figures and numbers in this document are generated directly from the model outputs.